EXPERIMENT -3

AIM- Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

OBJECTIVES

- To create relational tables for Department, Employee, and Project.
- To establish relationships between the tables using Primary Keys and Foreign Keys.
- To insert at least 30 employees, 5 departments, and 8 projects.
- To retrieve specific records using Selection.
- To retrieve specific attributes using Projection.
- To perform calculations using aggregate functions such as COUNT(), AVG(), MAX(), MIN(), and SUM().
- To group records using GROUP BY.
- To filter grouped results using HAVING.
- To categorize employee data using a CASE expression.
- To sort query results using ORDER BY.

THEORY

DATABASE SCHEMA

A database schema defines the logical structure of a database, including its tables, attributes, and relationships. In this experiment, the database consists of the Department, Employee, Project, and Employee_Project relations. The Department relation stores department information, the Employee relation stores employee details, the Project relation stores project information, and the Employee_Project relation associates employees with projects. The database contains 5 departments, 8 projects, and 30 employees.

SELECTION

Selection is used to retrieve records from a relation that satisfy a specified condition. In SQL, selection is generally performed using the WHERE clause. It filters the rows of a table according to the given condition.

PROJECTION

Projection is used to retrieve only the required attributes or columns from a relation. In SQL, projection is performed using the SELECT statement by specifying the required columns. It allows only the relevant attributes to be displayed in the result.

AGGREGATE FUNCTIONS

Aggregate functions perform calculations on a set of rows and return a single result. The aggregate functions used in this experiment are:

- COUNT() – Counts the number of rows.
- SUM() – Calculates the total of numeric values.
- AVG() – Calculates the average of numeric values.
- MAX() – Returns the maximum value.
- MIN() – Returns the minimum value.

GROUP BY

The GROUP BY clause is used to arrange rows into groups based on one or more columns. It is commonly used with aggregate functions to perform calculations separately for each group. In this experiment, it is used for department-wise analysis of employees and salaries.

HAVING

The HAVING clause is used to apply conditions to groups created using the GROUP BY clause. It is generally used with aggregate functions to filter grouped results based on a specified condition.

CASE EXPRESSION

The CASE expression is used to perform conditional operations in SQL. It evaluates specified conditions and returns a corresponding value when a condition is satisfied. In this experiment, it is used to categorize employees based on their salary.

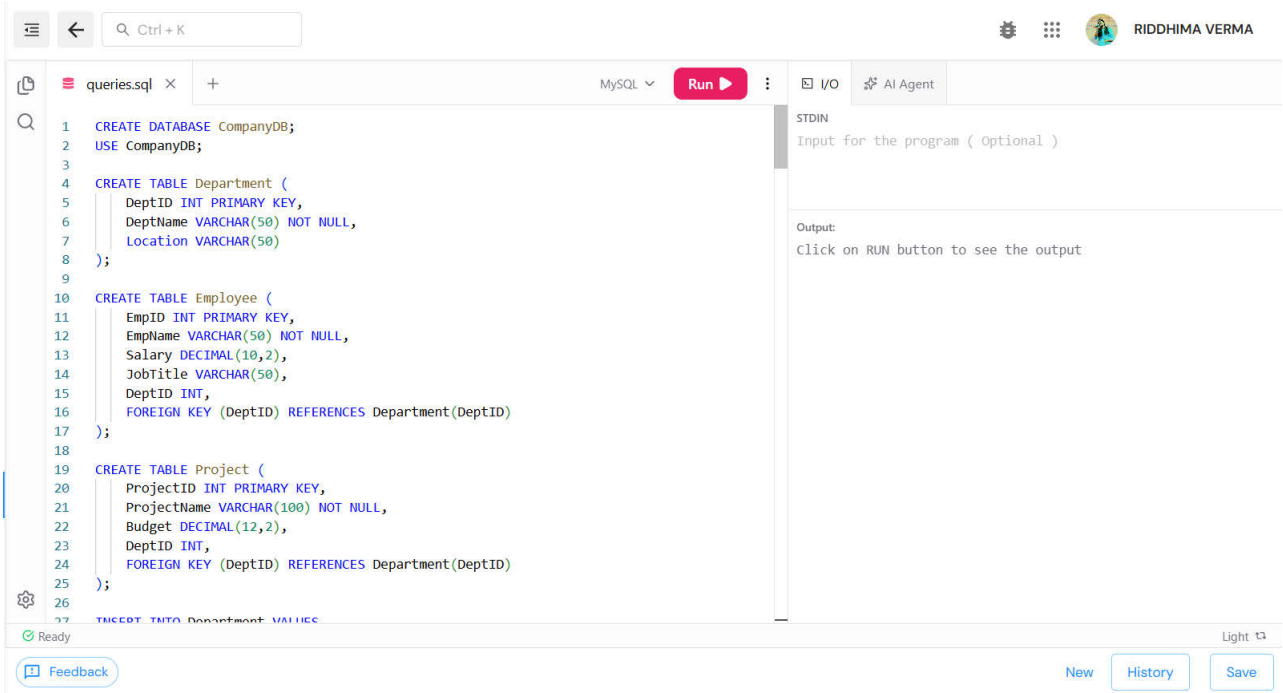
ORDER BY

The ORDER BY clause is used to sort the records returned by a query. The results can be arranged in ascending order using ASC or descending order using DESC. In this experiment, it is used to arrange employees according to their salary.

DATABASE RELATIONSHIPS

The database represents the relationship between employees, departments, and projects. A department can have multiple employees and projects. An employee can work on multiple projects, and a project can have multiple employees. The Employee_Project relation is used to represent the association between employees and projects.

SOURCE CODE-

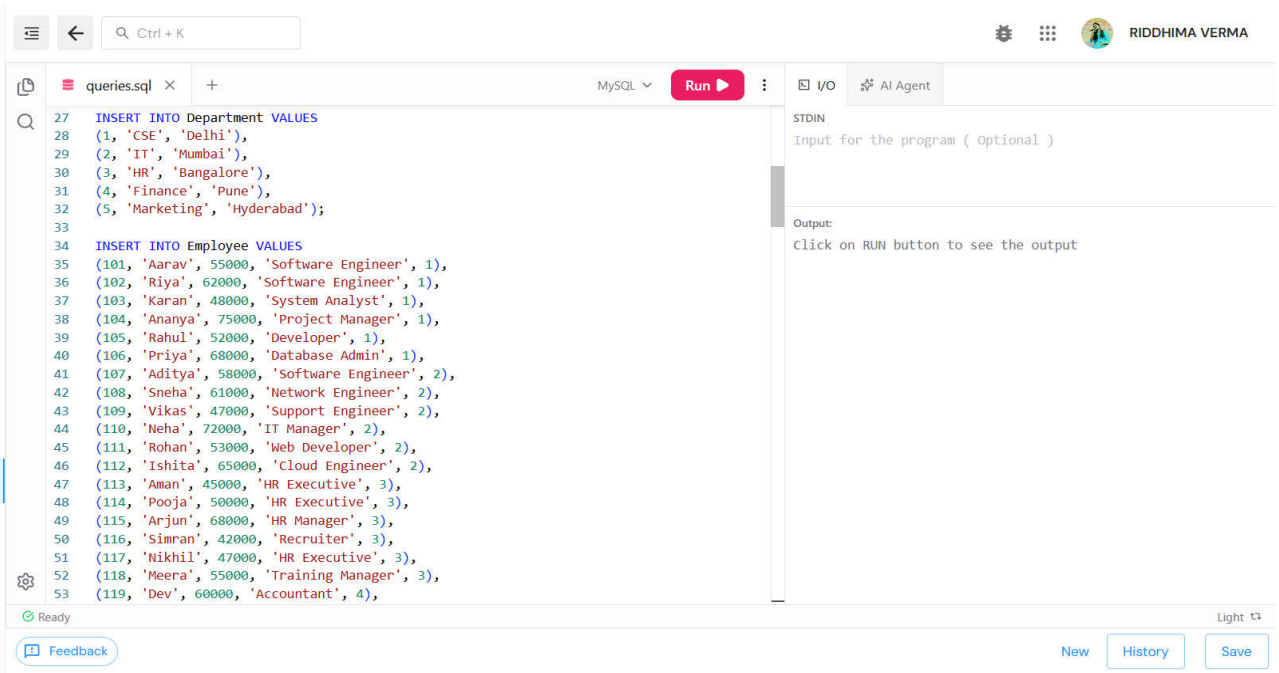



```sql
CREATE DATABASE CompanyDB;
USE CompanyDB;

CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50) NOT NULL,
    Location VARCHAR(50)
);

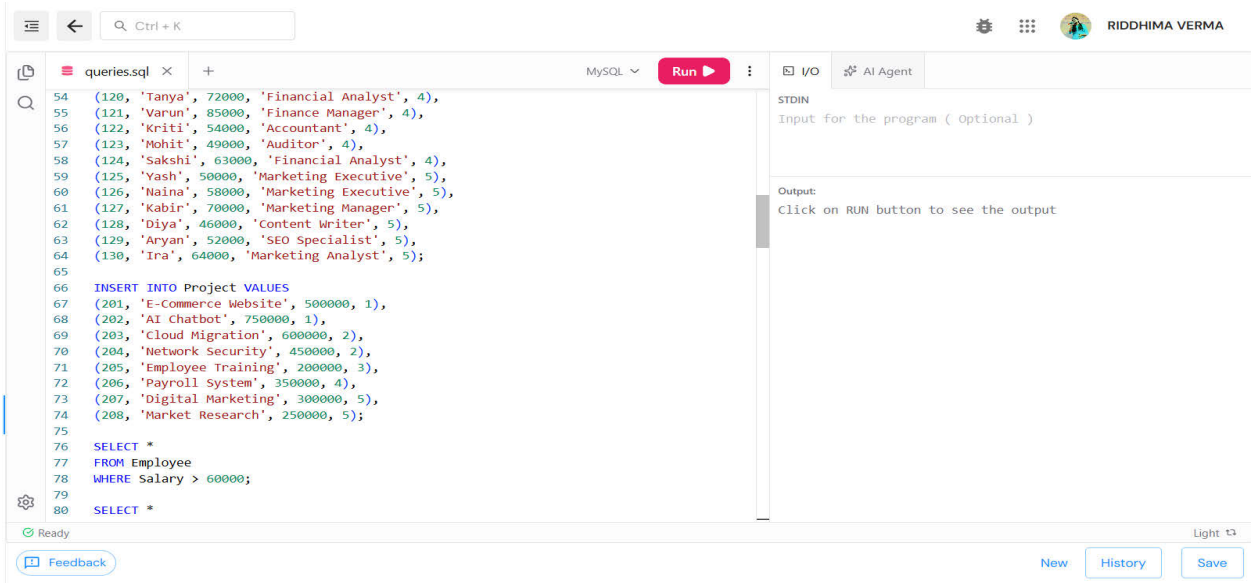
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50) NOT NULL,
    Salary DECIMAL(10,2),
    JobTitle VARCHAR(50),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);

CREATE TABLE Project (
    ProjectID INT PRIMARY KEY,
    ProjectName VARCHAR(100) NOT NULL,
    Budget DECIMAL(12,2),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
```




```sql
INSERT INTO Department VALUES
(1, 'CSE', 'Delhi'),
(2, 'IT', 'Mumbai'),
(3, 'HR', 'Bangalore'),
(4, 'Finance', 'Pune'),
(5, 'Marketing', 'Hyderabad');

INSERT INTO Employee VALUES
(101, 'Aarav', 55000, 'Software Engineer', 1),
(102, 'Riya', 62000, 'Software Engineer', 1),
(103, 'Karan', 48000, 'System Analyst', 1),
(104, 'Ananya', 75000, 'Project Manager', 1),
(105, 'Rahul', 52000, 'Developer', 1),
(106, 'Priya', 68000, 'Database Admin', 1),
(107, 'Aditya', 58000, 'Software Engineer', 2),
(108, 'Sneha', 61000, 'Network Engineer', 2),
(109, 'Vikas', 47000, 'Support Engineer', 2),
(110, 'Neha', 72000, 'IT Manager', 2),
(111, 'Rohan', 53000, 'Web Developer', 2),
(112, 'Ishita', 65000, 'Cloud Engineer', 2),
(113, 'Aman', 45000, 'HR Executive', 3),
(114, 'Pooja', 50000, 'HR Executive', 3),
(115, 'Arjun', 68000, 'HR Manager', 3),
(116, 'Simran', 42000, 'Recruiter', 3),
(117, 'Nikhil', 47000, 'HR Executive', 3),
(118, 'Meera', 55000, 'Training Manager', 3),
(119, 'Dev', 60000, 'Accountant', 4),
```

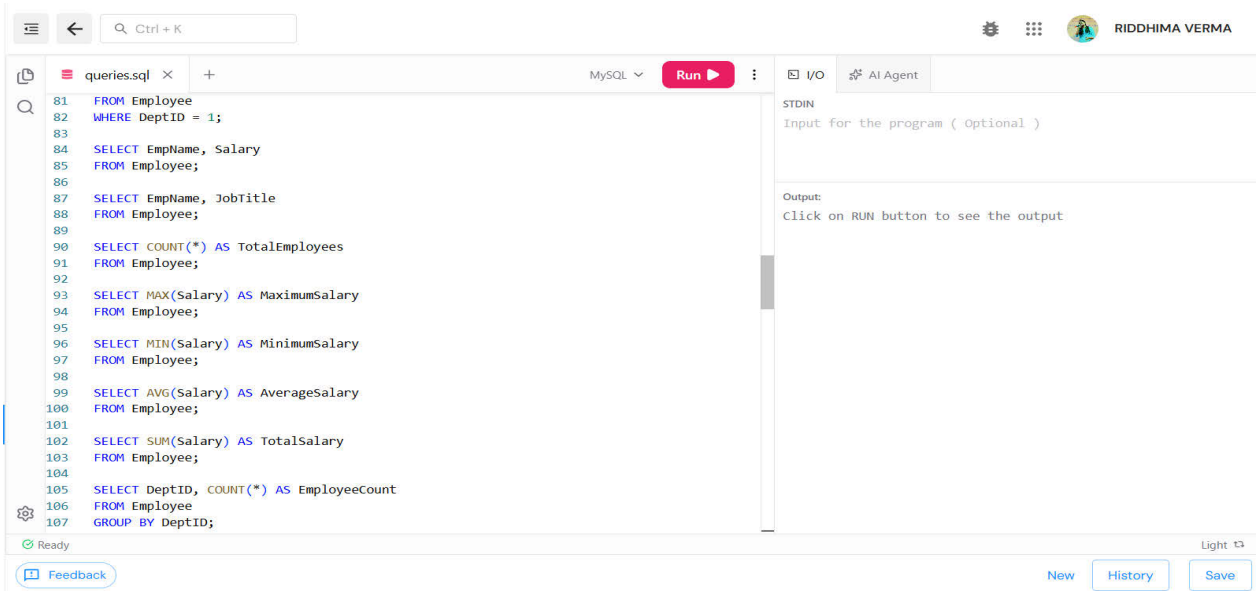



```
(120, 'Tanya', 72000, 'Financial Analyst', 4),
(121, 'Varun', 85000, 'Finance Manager', 4),
(122, 'Kriti', 54000, 'Accountant', 4),
(123, 'Mohit', 49000, 'Auditor', 4),
(124, 'Sakshi', 63000, 'Financial Analyst', 4),
(125, 'Yash', 50000, 'Marketing Executive', 5),
(126, 'Naina', 58000, 'Marketing Executive', 5),
(127, 'Kabir', 70000, 'Marketing Manager', 5),
(128, 'Diya', 46000, 'Content Writer', 5),
(129, 'Aryan', 52000, 'SEO Specialist', 5),
(130, 'Ira', 64000, 'Marketing Analyst', 5);

INSERT INTO Project VALUES
(201, 'E-Commerce Website', 500000, 1),
(202, 'AI Chatbot', 750000, 1),
(203, 'Cloud Migration', 600000, 2),
(204, 'Network Security', 450000, 2),
(205, 'Employee Training', 200000, 3),
(206, 'Payroll System', 350000, 4),
(207, 'Digital Marketing', 300000, 5),
(208, 'Market Research', 250000, 5);

SELECT *
FROM Employee
WHERE Salary > 60000;

SELECT *
```




```
FROM Employee
WHERE DeptID = 1;

SELECT EmpName, Salary
FROM Employee;

SELECT EmpName, JobTitle
FROM Employee;

SELECT COUNT(*) AS TotalEmployees
FROM Employee;

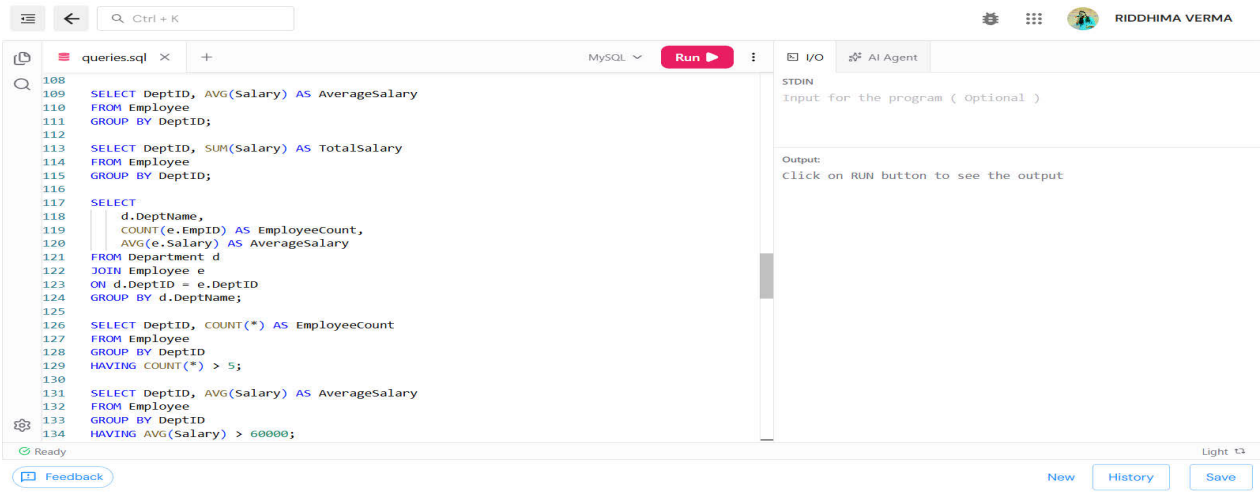
SELECT MAX(Salary) AS MaximumSalary
FROM Employee;

SELECT MIN(Salary) AS MinimumSalary
FROM Employee;

SELECT AVG(Salary) AS AverageSalary
FROM Employee;

SELECT SUM(Salary) AS TotalSalary
FROM Employee;

SELECT DeptID, COUNT(*) AS EmployeeCount
FROM Employee
GROUP BY DeptID;
```



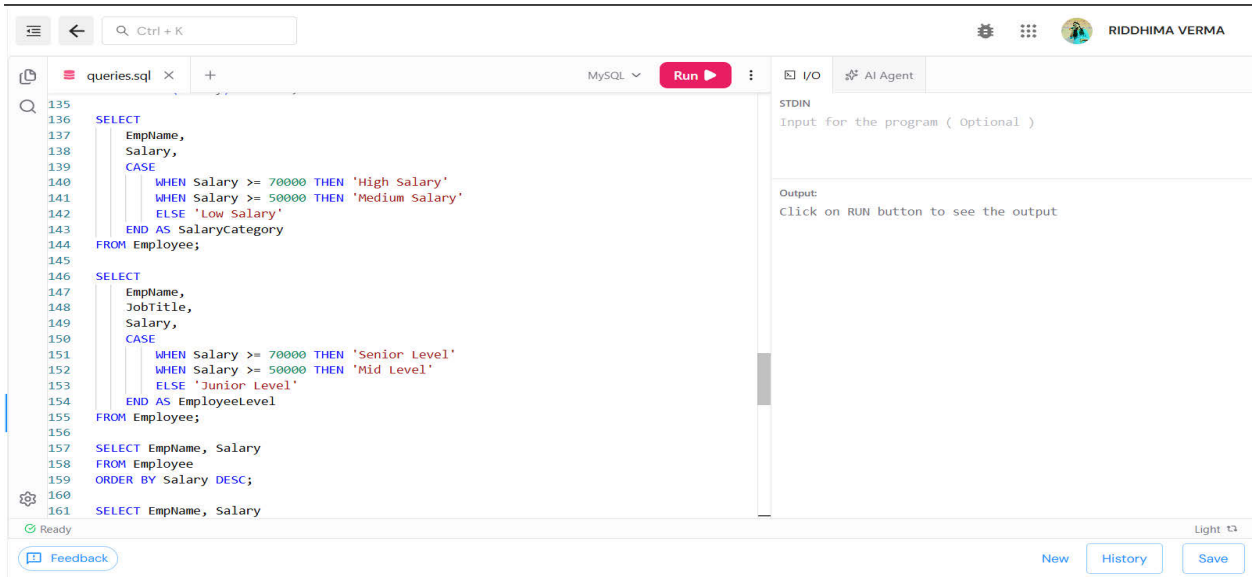

```
SELECT DeptID, AVG(Salary) AS AverageSalary
FROM Employee
GROUP BY DeptID;

SELECT DeptID, SUM(Salary) AS TotalSalary
FROM Employee
GROUP BY DeptID;

SELECT
    d.DeptName,
    COUNT(e.EmpID) AS EmployeeCount,
    AVG(e.Salary) AS AverageSalary
FROM Department d
JOIN Employee e
ON d.DeptID = e.DeptID
GROUP BY d.DeptName;

SELECT DeptID, COUNT(*) AS EmployeeCount
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 5;

SELECT DeptID, AVG(Salary) AS AverageSalary
FROM Employee
GROUP BY DeptID
HAVING AVG(Salary) > 60000;
```

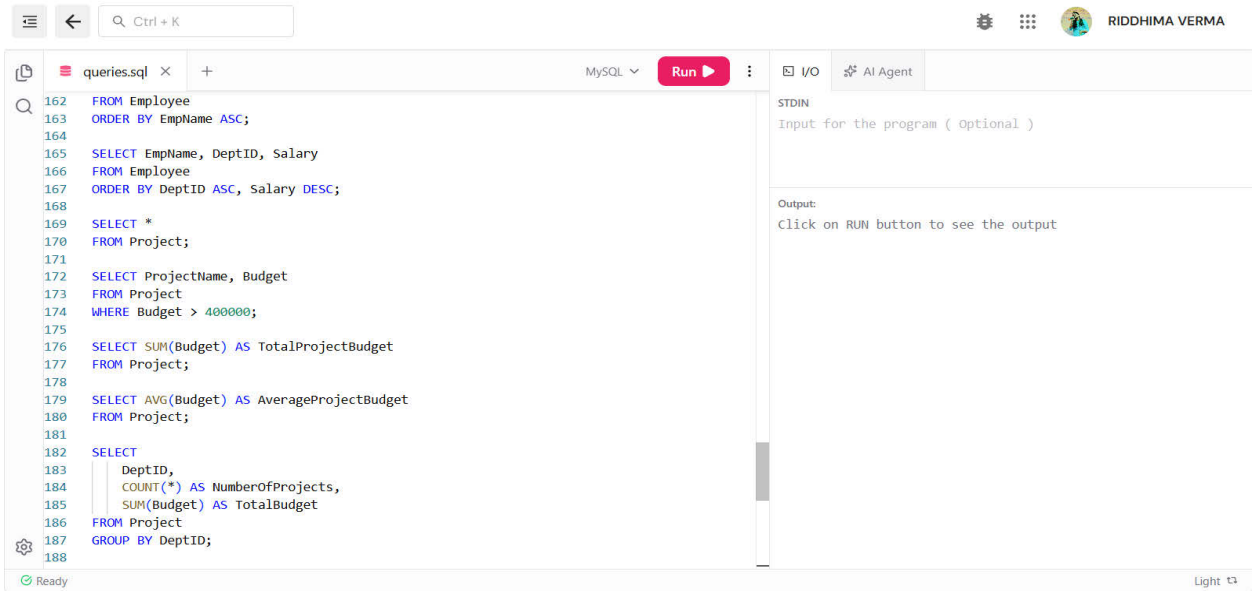



```sql
SELECT
    EmpName,
    Salary,
    CASE
        WHEN Salary >= 70000 THEN 'High Salary'
        WHEN Salary >= 50000 THEN 'Medium Salary'
        ELSE 'Low Salary'
    END AS SalaryCategory
FROM Employee;

SELECT
    EmpName,
    JobTitle,
    Salary,
    CASE
        WHEN Salary >= 70000 THEN 'Senior Level'
        WHEN Salary >= 50000 THEN 'Mid Level'
        ELSE 'Junior Level'
    END AS EmployeeLevel
FROM Employee;

SELECT EmpName, Salary
FROM Employee
ORDER BY Salary DESC;

SELECT EmpName, Salary
```




```sql
FROM Employee
ORDER BY EmpName ASC;

SELECT EmpName, DeptID, Salary
FROM Employee
ORDER BY DeptID ASC, Salary DESC;

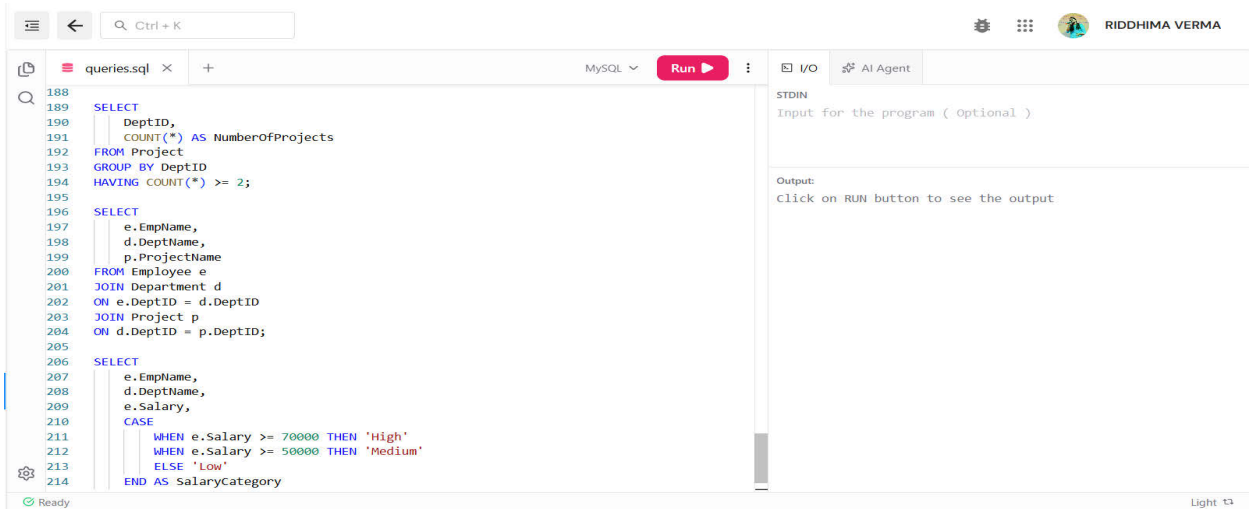
SELECT *
FROM Project;

SELECT ProjectName, Budget
FROM Project
WHERE Budget > 400000;

SELECT SUM(Budget) AS TotalProjectBudget
FROM Project;

SELECT AVG(Budget) AS AverageProjectBudget
FROM Project;

SELECT
    DeptID,
    COUNT(*) AS NumberOfProjects,
    SUM(Budget) AS TotalBudget
FROM Project
GROUP BY DeptID;
```




```sql
SELECT
    DeptID,
    COUNT(*) AS NumberOfProjects
FROM Project
GROUP BY DeptID
HAVING COUNT(*) >= 2;

SELECT
    e.EmpName,
    d.DeptName,
    p.ProjectName
FROM Employee e
JOIN Department d
ON e.DeptID = d.DeptID
JOIN Project p
ON d.DeptID = p.DeptID;

SELECT
    e.EmpName,
    d.DeptName,
    e.Salary,
    CASE
        WHEN e.Salary >= 70000 THEN 'High'
        WHEN e.Salary >= 50000 THEN 'Medium'
        ELSE 'Low'
    END AS SalaryCategory
```

```sql
GROUP BY DeptID
HAVING COUNT(*) >= 2;

SELECT
    e.EmpName,
    d.DeptName,
    p.ProjectName
FROM Employee e
JOIN Department d
ON e.DeptID = d.DeptID
JOIN Project p
ON d.DeptID = p.DeptID;

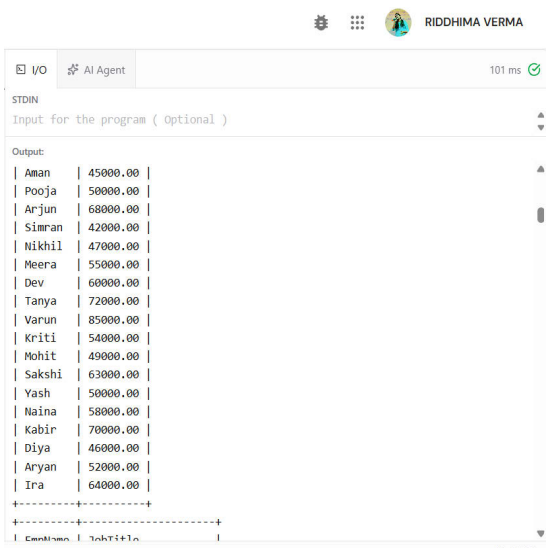
SELECT
    e.EmpName,
    d.DeptName,
    e.Salary,
    CASE
        WHEN e.Salary >= 70000 THEN 'High'
        WHEN e.Salary >= 50000 THEN 'Medium'
        ELSE 'Low'
    END AS SalaryCategory
FROM Employee e
JOIN Department d
ON e.DeptID = d.DeptID
WHERE e.Salary >= 45000
ORDER BY e.Salary DESC;
```

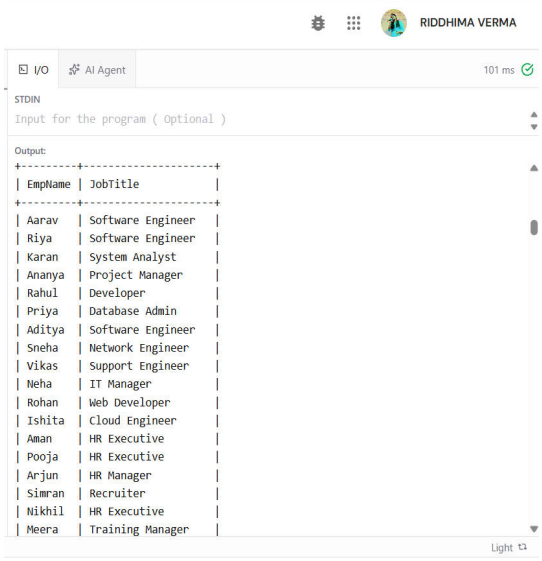
OUTPUT

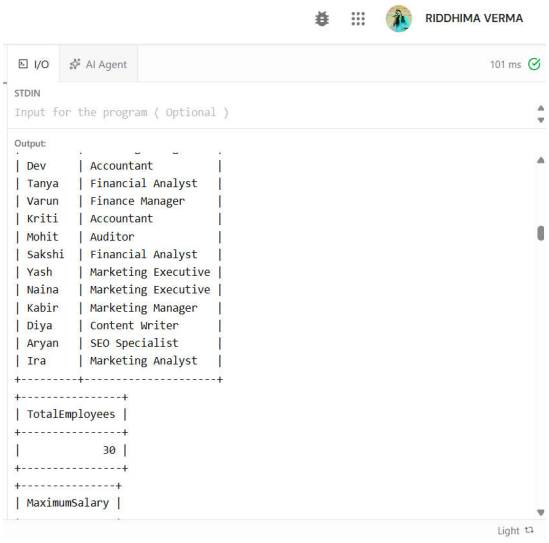
```
| EmpID | EmpName | Salary   | JobTitle          | DeptID |
|   102 | Riya    | 62000.00 | Software Engineer |      1 |
|   104 | Ananya  | 75000.00 | Project Manager   |      1 |
|   106 | Priya   | 68000.00 | Database Admin    |      1 |
|   108 | Sneha   | 61000.00 | Network Engineer  |      2 |
|   110 | Neha    | 72000.00 | IT Manager        |      2 |
|   112 | Ishita  | 65000.00 | Cloud Engineer    |      2 |
|   115 | Arjun   | 68000.00 | HR Manager        |      3 |
|   120 | Tanya   | 72000.00 | Financial Analyst |      4 |
|   121 | Varun   | 85000.00 | Finance Manager   |      4 |
|   124 | Sakshi  | 63000.00 | Financial Analyst |      4 |
|   127 | Kabir   | 70000.00 | Marketing Manager |      5 |
|   130 | Ira     | 64000.00 | Marketing Analyst |      5 |

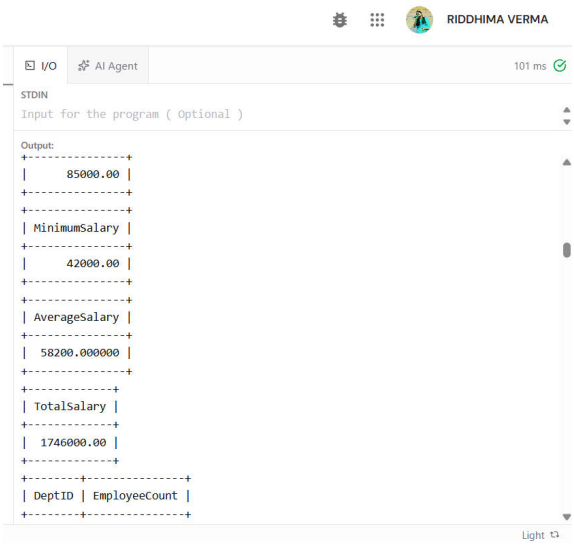
| EmpID | EmpName | Salary   | JobTitle          | DeptID |
|   101 | Aarav   | 55000.00 | Software Engineer |      1 |
|   102 | Riya    | 62000.00 | Software Engineer |      1 |
|   103 | Karan   | 48000.00 | System Analyst    |      1 |
|   104 | Ananya  | 75000.00 | Project Manager   |      1 |
|   105 | Rahul   | 52000.00 | Developer         |      1 |
|   106 | Priya   | 68000.00 | Database Admin    |      1 |

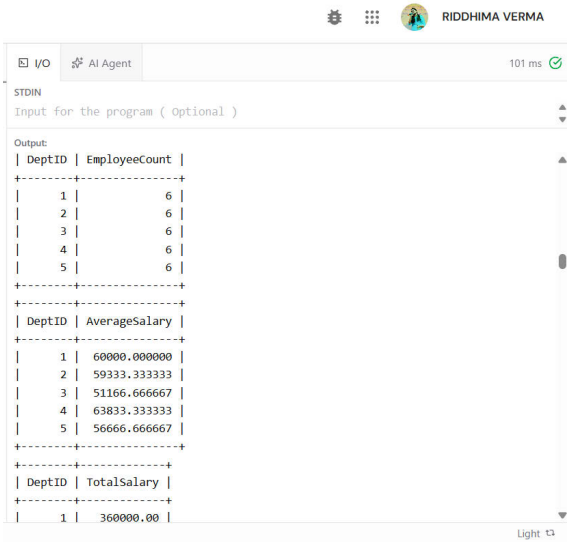
| EmpName | Salary   |
| Aarav   | 55000.00 |
| Riya    | 62000.00 |
| Karan   | 48000.00 |
| Ananya  | 75000.00 |
| Rahul   | 52000.00 |
| Priya   | 68000.00 |
| Aditya  | 58000.00 |
| Sneha   | 61000.00 |
| Vikas   | 47000.00 |
| Neha    | 72000.00 |
| Rohan   | 53000.00 |
| Ishita  | 65000.00 |
```

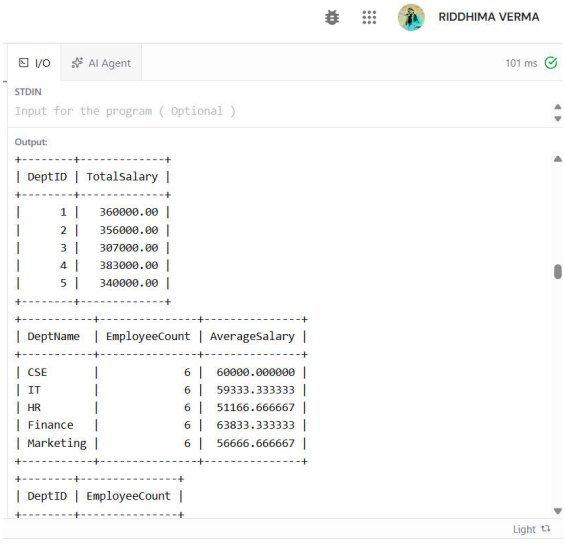

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Aman	45000.00
Pooja	50000.00
Arjun	68000.00
Simran	42000.00
Nikhil	47000.00
Meera	55000.00
Dev	60000.00
Tanya	72000.00
Varun	85000.00
Kriti	54000.00
Mohit	49000.00
Sakshi	63000.00
Yash	50000.00
Naina	58000.00
Kabir	70000.00
Diya	46000.00
Aryan	52000.00
Ira	64000.00
Light

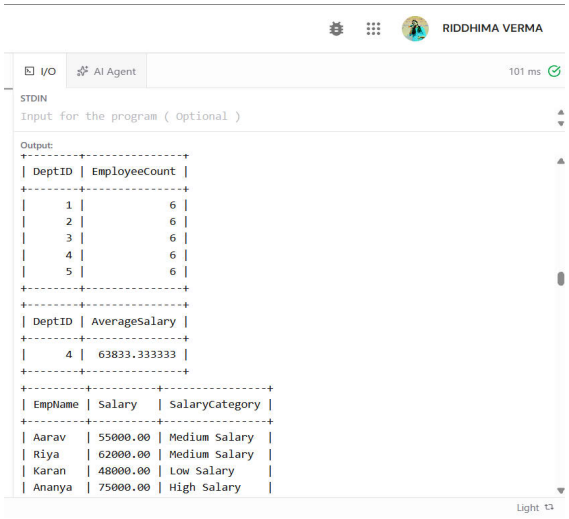

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
EmpName	JobTitle
Aarav	Software Engineer
Riya	Software Engineer
Karan	System Analyst
Ananya	Project Manager
Rahul	Developer
Priya	Database Admin
Aditya	Software Engineer
Sneha	Network Engineer
Vikas	Support Engineer
Neha	IT Manager
Rohan	Web Developer
Ishita	Cloud Engineer
Aman	HR Executive
Pooja	HR Executive
Arjun	HR Manager
Simran	Recruiter
Nikhil	HR Executive
Meera	Training Manager
Light

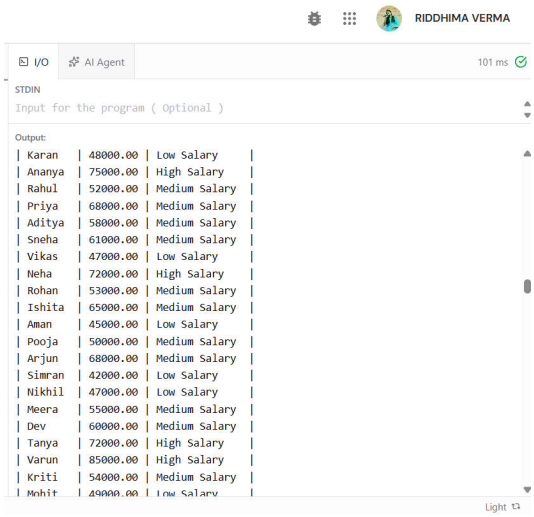

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Dev	Accountant
Tanya	Financial Analyst
Varun	Finance Manager
Kriti	Accountant
Mohit	Auditor
Sakshi	Financial Analyst
Yash	Marketing Executive
Naina	Marketing Executive
Kabir	Marketing Manager
Diya	Content Writer
Aryan	SEO Specialist
Ira	Marketing Analyst
TotalEmployees
30
MaximumSalary
Light

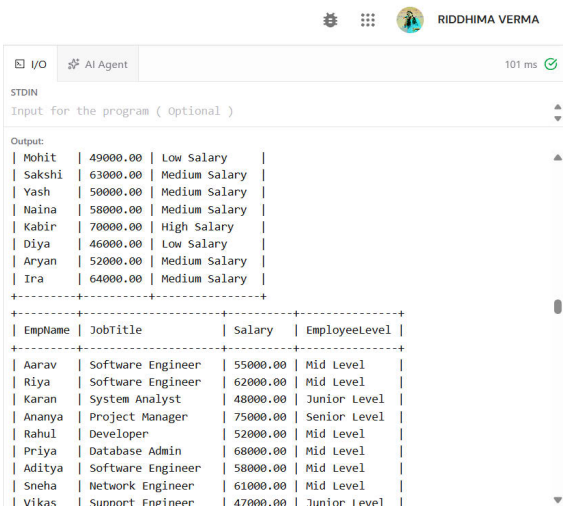

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
| 85000.00 |
| MinimumSalary |
| 42000.00 |
| AverageSalary |
| 58200.000000 |
| TotalSalary |
| 1746000.00 |
| DeptID | EmployeeCount |
Light

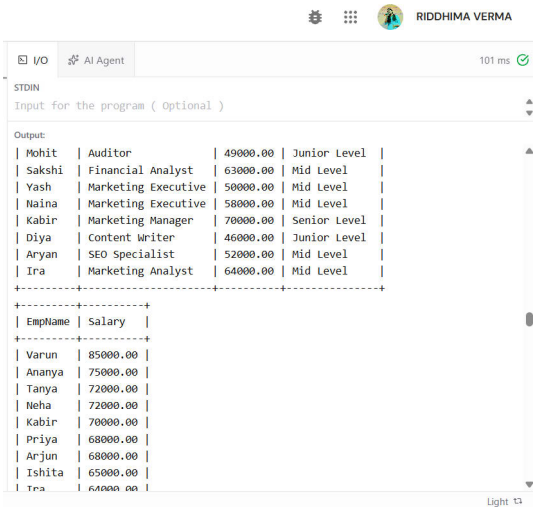

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
DeptID	EmployeeCount
1	6
2	6
3	6
4	6
5	6
DeptID	AverageSalary
1	60000.000000
2	59333.333333
3	51166.666667
4	63833.333333
5	56666.666667
DeptID	TotalSalary
1	360000.00
Light

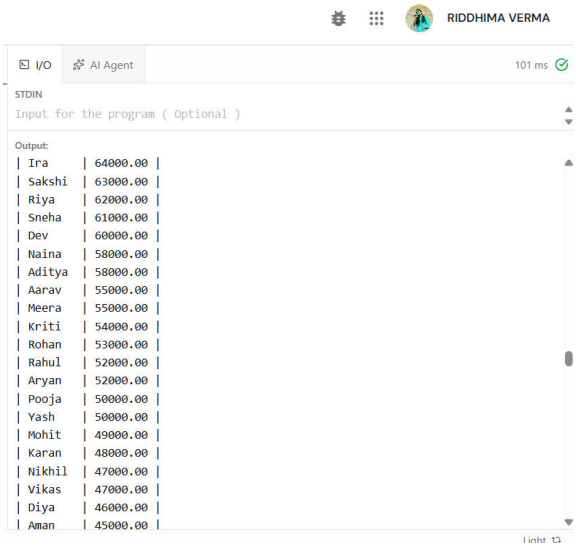

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
DeptID	TotalSalary
1	360000.00
2	356000.00
3	307000.00
4	383000.00
5	340000.00
DeptName	EmployeeCount	AverageSalary
CSE	6	60000.000000
IT	6	59333.333333
HR	6	51166.666667
Finance	6	63833.333333
Marketing	6	56666.666667
DeptID	EmployeeCount
Light

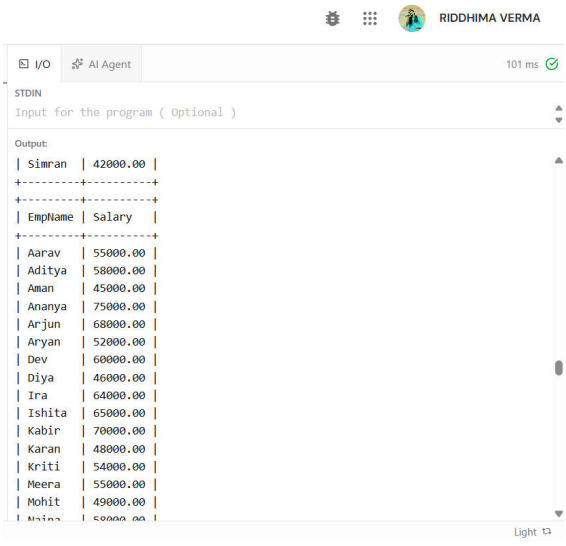

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
+--------+---------------+
| DeptID | EmployeeCount |
+--------+---------------+
1	6
2	6
3	6
4	6
5	6
+--------+---------------+
+--------+---------------+
| DeptID | AverageSalary |
+--------+---------------+
| 4 | 63833.333333 |
+--------+---------------+
+---------+---------+----------------+
| EmpName | Salary | SalaryCategory |
+---------+---------+----------------+
Aarav	55000.00	Medium Salary
Riya	62000.00	Medium Salary
Karan	48000.00	Low Salary
Ananya	75000.00	High Salary
Light

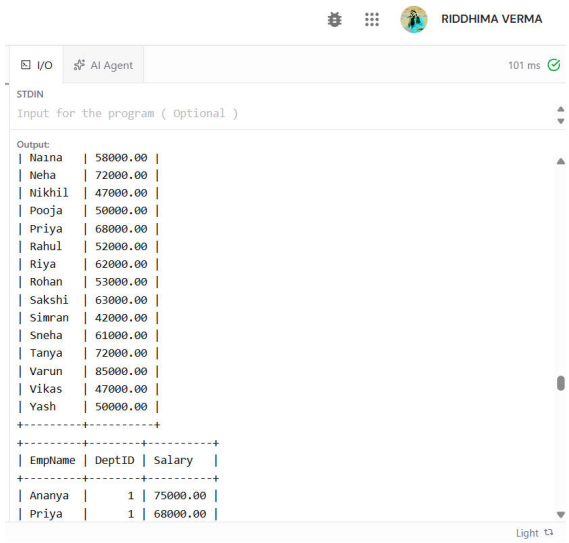

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Karan	48000.00	Low Salary
Ananya	75000.00	High Salary
Rahul	52000.00	Medium Salary
Priya	68000.00	Medium Salary
Aditya	58000.00	Medium Salary
Sneha	61000.00	Medium Salary
Vikas	47000.00	Low Salary
Neha	72000.00	High Salary
Rohan	53000.00	Medium Salary
Ishita	65000.00	Medium Salary
Aman	45000.00	Low Salary
Pooja	50000.00	Medium Salary
Arjun	68000.00	Medium Salary
Simran	42000.00	Low Salary
Nikhil	47000.00	Low Salary
Meera	55000.00	Medium Salary
Dev	60000.00	Medium Salary
Tanya	72000.00	High Salary
Varun	85000.00	High Salary
Kriti	54000.00	Medium Salary
Light

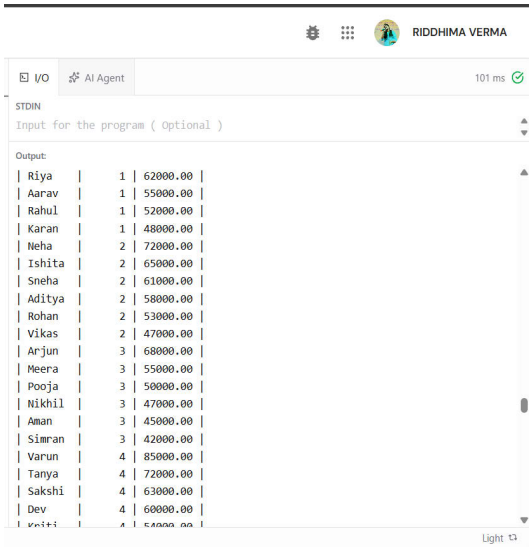

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Mohit	49000.00	Low Salary
Sakshi	63000.00	Medium Salary
Yash	50000.00	Medium Salary
Naina	58000.00	Medium Salary
Kabir	70000.00	High Salary
Diya	46000.00	Low Salary
Aryan	52000.00	Medium Salary
Ira	64000.00	Medium Salary
+---------+---------+----------------+
+---------+--------------------+---------+---------------+
| EmpName | JobTitle | Salary | EmployeeLevel |
+---------+--------------------+---------+---------------+
Aarav	Software Engineer	55000.00	Mid Level
Riya	Software Engineer	62000.00	Mid Level
Karan	System Analyst	48000.00	Junior Level
Ananya	Project Manager	75000.00	Senior Level
Rahul	Developer	52000.00	Mid Level
Priya	Database Admin	68000.00	Mid Level
Aditya	Software Engineer	58000.00	Mid Level
Sneha	Network Engineer	61000.00	Mid Level
Vikas	Support Engineer	47000.00	Junior Level

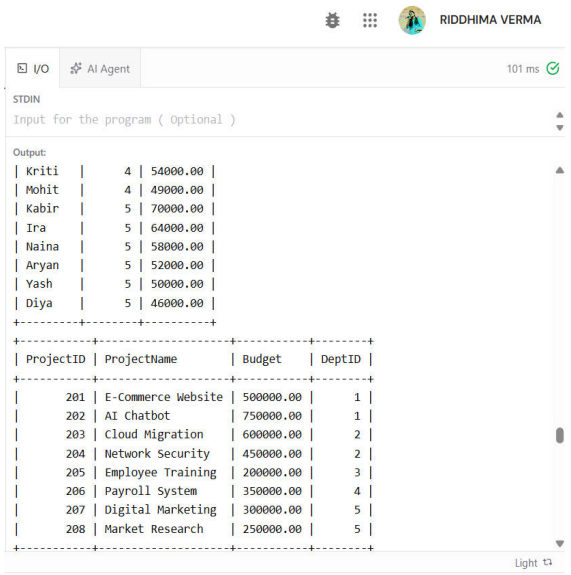

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Mohit	Auditor	49000.00	Junior Level
Sakshi	Financial Analyst	63000.00	Mid Level
Yash	Marketing Executive	50000.00	Mid Level
Naina	Marketing Executive	58000.00	Mid Level
Kabir	Marketing Manager	70000.00	Senior Level
Diya	Content Writer	46000.00	Junior Level
Aryan	SEO Specialist	52000.00	Mid Level
Ira	Marketing Analyst	64000.00	Mid Level
+---------+--------------------+---------+---------------+
+---------+---------+
| EmpName | Salary |
+---------+---------+
Varun	85000.00
Ananya	75000.00
Tanya	72000.00
Neha	72000.00
Kabir	70000.00
Priya	68000.00
Arjun	68000.00
Ishita	65000.00
Light

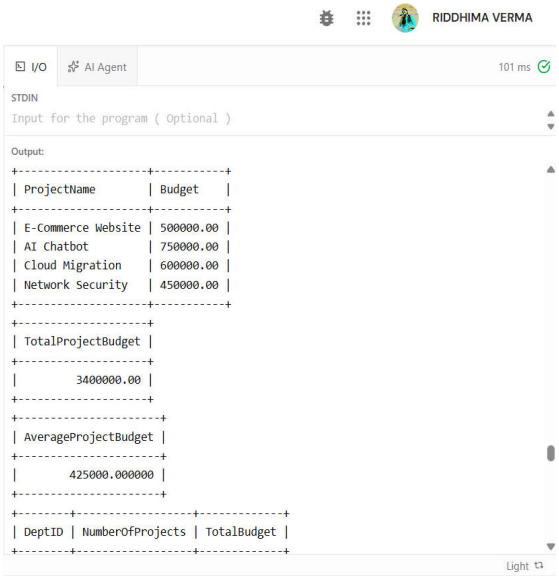

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Ira	64000.00
Sakshi	63000.00
Riya	62000.00
Sneha	61000.00
Dev	60000.00
Naina	58000.00
Aditya	58000.00
Aarav	55000.00
Meera	55000.00
Kriti	54000.00
Rohan	53000.00
Rahul	52000.00
Aryan	52000.00
Pooja	50000.00
Yash	50000.00
Mohit	49000.00
Karan	48000.00
Nikhil	47000.00
Vikas	47000.00
Diya	46000.00
Aman	45000.00
Light

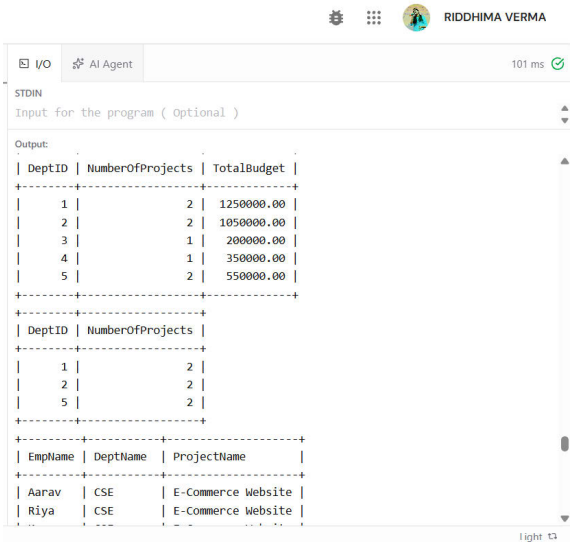

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
| Simran | 42000.00 |
+---------+---------+
+---------+---------+
| EmpName | Salary |
+---------+---------+
Aarav	55000.00
Aditya	58000.00
Aman	45000.00
Ananya	75000.00
Arjun	68000.00
Aryan	52000.00
Dev	60000.00
Diya	46000.00
Ira	64000.00
Ishita	65000.00
Kabir	70000.00
Karan	48000.00
Kriti	54000.00
Meera	55000.00
Mohit	49000.00
Light

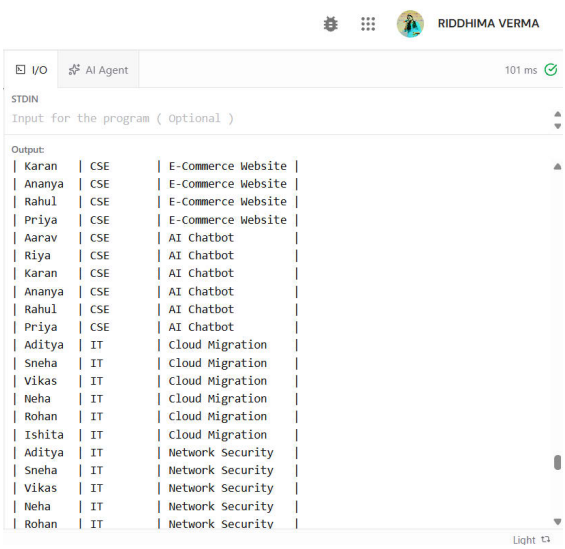

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Naina	58000.00
Neha	72000.00
Nikhil	47000.00
Pooja	50000.00
Priya	68000.00
Rahul	52000.00
Riya	62000.00
Rohan	53000.00
Sakshi	63000.00
Simran	42000.00
Sneha	61000.00
Tanya	72000.00
Varun	85000.00
Vikas	47000.00
Yash	50000.00
+---------+----------+
+---------+--------+----------+
| EmpName | DeptID | Salary |
+---------+--------+----------+
| Ananya | 1 | 75000.00 |
| Priya | 1 | 68000.00 |
Light

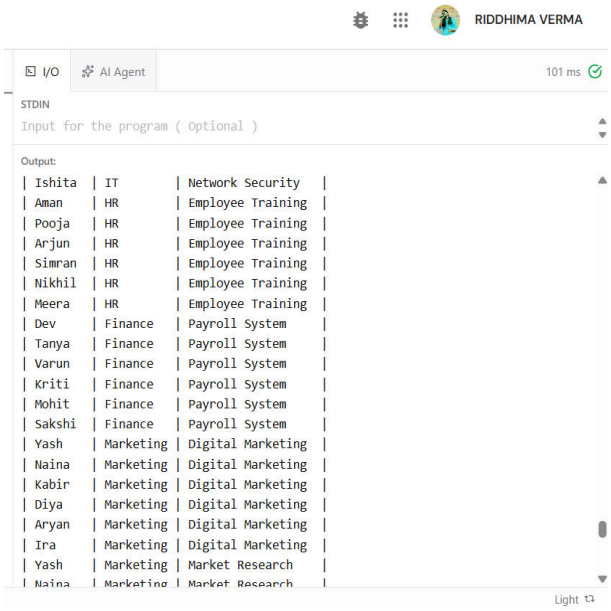

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Riya	1	62000.00
Aarav	1	55000.00
Rahul	1	52000.00
Karan	1	48000.00
Neha	2	72000.00
Ishita	2	65000.00
Sneha	2	61000.00
Aditya	2	58000.00
Rohan	2	53000.00
Vikas	2	47000.00
Arjun	3	68000.00
Meera	3	55000.00
Pooja	3	50000.00
Nikhil	3	47000.00
Aman	3	45000.00
Simran	3	42000.00
Varun	4	85000.00
Tanya	4	72000.00
Sakshi	4	63000.00
Dev	4	60000.00
Light

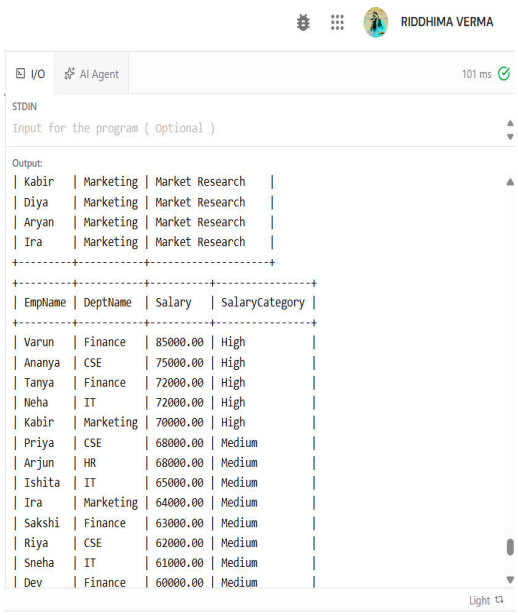

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Kriti	4	54000.00
Mohit	4	49000.00
Kabir	5	70000.00
Ira	5	64000.00
Naina	5	58000.00
Aryan	5	52000.00
Yash	5	50000.00
Diya	5	46000.00
+---------+-------+---------+
+-----------+--------------------+----------+--------+
| ProjectID | ProjectName | Budget | DeptID |
+-----------+--------------------+----------+--------+
201	E-Commerce Website	500000.00	1
202	AI Chatbot	750000.00	1
203	Cloud Migration	600000.00	2
204	Network Security	450000.00	2
205	Employee Training	200000.00	3
206	Payroll System	350000.00	4
207	Digital Marketing	300000.00	5
208	Market Research	250000.00	5
+-----------+--------------------+----------+--------+
Light

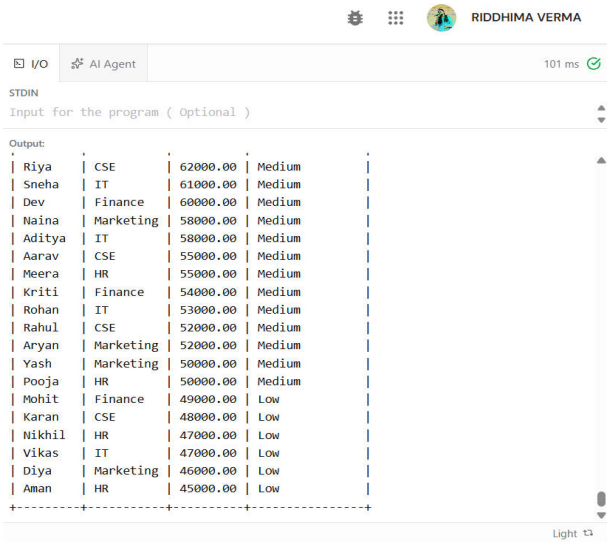

RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
+--------------------+----------+
| ProjectName | Budget |
+--------------------+----------+
E-Commerce Website	500000.00
AI Chatbot	750000.00
Cloud Migration	600000.00
Network Security	450000.00
+--------------------+----------+
+--------------------+
| TotalProjectBudget |
+--------------------+
| 3400000.00 |
+--------------------+
+----------------------+
| AverageProjectBudget |
+----------------------+
| 425000.000000 |
+----------------------+
+--------+-----------------+-------------+
| DeptID | NumberOfProjects | TotalBudget |
+--------+-----------------+-------------+
Light


RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
| DeptID | NumberOfProjects | TotalBudget |
+--------+-----------------+-------------+
1	2	1250000.00
2	2	1050000.00
3	1	200000.00
4	1	350000.00
5	2	550000.00
+--------+-----------------+-------------+
+--------+-----------------+
| DeptID | NumberOfProjects |
+--------+-----------------+
1	2
2	2
5	2
+--------+-----------------+
+---------+----------+--------------------+
| EmpName | DeptName | ProjectName |
+---------+----------+--------------------+
| Aarav | CSE | E-Commerce Website |
| Riya | CSE | E-Commerce Website |
Light


RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Karan	CSE	E-Commerce Website
Ananya	CSE	E-Commerce Website
Rahul	CSE	E-Commerce Website
Priya	CSE	E-Commerce Website
Aarav	CSE	AI Chatbot
Riya	CSE	AI Chatbot
Karan	CSE	AI Chatbot
Ananya	CSE	AI Chatbot
Rahul	CSE	AI Chatbot
Priya	CSE	AI Chatbot
Aditya	IT	Cloud Migration
Sneha	IT	Cloud Migration
Vikas	IT	Cloud Migration
Neha	IT	Cloud Migration
Rohan	IT	Cloud Migration
Ishita	IT	Cloud Migration
Aditya	IT	Network Security
Sneha	IT	Network Security
Vikas	IT	Network Security
Neha	IT	Network Security
Light


RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Ishita	IT	Network Security
Aman	HR	Employee Training
Pooja	HR	Employee Training
Arjun	HR	Employee Training
Simran	HR	Employee Training
Nikhil	HR	Employee Training
Meera	HR	Employee Training
Dev	Finance	Payroll System
Tanya	Finance	Payroll System
Varun	Finance	Payroll System
Kriti	Finance	Payroll System
Mohit	Finance	Payroll System
Sakshi	Finance	Payroll System
Yash	Marketing	Digital Marketing
Naina	Marketing	Digital Marketing
Kabir	Marketing	Digital Marketing
Diya	Marketing	Digital Marketing
Aryan	Marketing	Digital Marketing
Ira	Marketing	Digital Marketing
Yash	Marketing	Market Research
Light


RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Kabir	Marketing	Market Research
Diya	Marketing	Market Research
Aryan	Marketing	Market Research
Ira	Marketing	Market Research
EmpName	DeptName	Salary	SalaryCategory
Varun	Finance	85000.00	High
Ananya	CSE	75000.00	High
Tanya	Finance	72000.00	High
Neha	IT	72000.00	High
Kabir	Marketing	70000.00	High
Priya	CSE	68000.00	Medium
Arjun	HR	68000.00	Medium
Ishita	IT	65000.00	Medium
Ira	Marketing	64000.00	Medium
Sakshi	Finance	63000.00	Medium
Riya	CSE	62000.00	Medium
Sneha	IT	61000.00	Medium
Dev	Finance	60000.00	Medium
Light


RIDDHIMA VERMA
I/O
AI Agent
101 ms
STDIN
Input for the program (Optional)
Output:
Riya	CSE	62000.00	Medium
Sneha	IT	61000.00	Medium
Dev	Finance	60000.00	Medium
Naina	Marketing	58000.00	Medium
Aditya	IT	58000.00	Medium
Aarav	CSE	55000.00	Medium
Meera	HR	55000.00	Medium
Kriti	Finance	54000.00	Medium
Rohan	IT	53000.00	Medium
Rahul	CSE	52000.00	Medium
Aryan	Marketing	52000.00	Medium
Yash	Marketing	50000.00	Medium
Pooja	HR	50000.00	Medium
Mohit	Finance	49000.00	Low
Karan	CSE	48000.00	Low
Nikhil	HR	47000.00	Low
Vikas	IT	47000.00	Low
Diya	Marketing	46000.00	Low
Aman	HR	45000.00	Low
Light

OBSERVATION/OUTCOME

1. Employee, Department, and Project tables were successfully created.
2. Data for 30 employees, 5 departments, and 8 projects was successfully inserted.
3. Selection and projection queries successfully retrieved the required records and attributes.
4. Aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() were successfully executed.
5. GROUP BY and HAVING were used to group and filter grouped data.
6. CASE expression successfully categorized employees based on salary.
7. ORDER BY successfully sorted employee records according to the specified conditions.
8. Queries using joins successfully retrieved related data from multiple tables.